

```

1.WITH salary_data AS
(
    SELECT
        emp_id,
        salary,
        salary_date,
        LAG(salary) OVER (
            PARTITION BY emp_id
            ORDER BY salary_date
        ) AS previous_salary
    FROM salary_history
)
SELECT
    emp_id,
    salary AS latest_salary,
    previous_salary,
    (salary - previous_salary) * 100.0 / previous_salary AS increase_percent
FROM salary_data

```

```

2.WITH cte AS
(
    SELECT
        product_id,
        price,
        price_date,
        LAG(price) OVER
        (
            PARTITION BY product_id
            ORDER BY price_date
        ) AS previous_price
    FROM product_price
)

```

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3.SELECT customer_id
FROM orders
WHERE YEAR(order_date) = 2026
GROUP BY customer_id
HAVING COUNT(DISTINCT MONTH(order_date)) = 12

```

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4.WITH CTE AS
(
    SELECT SALE_AMOUNT
    FROM SALES
    GROUP BY SALE_AMOUNT
    HAVING COUNT(*) > 1
)
SELECT *
FROM SALES
WHERE SALE_AMOUNT IN
(
    SELECT SALE_AMOUNT
    FROM CTE
)

```

```

5.WITH CTE AS
(
    SELECT
        C.CUST_NAME,
        SUM(O.AMOUNT) AS TOTAL_SPENT
    FROM CUSTOMERS C
    JOIN ORDERS O
        ON C.CUST_ID = O.CUST_ID
    GROUP BY C.CUST_NAME
),

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RANK_CUST AS
(
    SELECT
        CUST_NAME,
        TOTAL_SPENT,
        DENSE_RANK() OVER(ORDER BY TOTAL_SPENT DESC) AS RANK
    FROM CTE
)
SELECT
    CUST_NAME,
    TOTAL_SPENT,
    RANK
FROM RANK_CUST
WHERE RANK <= 5

```

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6.WITH CTE AS
(
    SELECT
        DATEPART(HOUR, order_time) AS sale_hour,
        SUM(amount) AS total_revenue
    FROM sales
    GROUP BY DATEPART(HOUR, order_time)
)
SELECT TOP 1
    sale_hour,
    total_revenue
FROM CTE
ORDER BY total_revenue DESC

```

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7.WITH CTE AS
(
    SELECT
        P.CATEGORY_ID,
        P.PRODUCT_NAME,
        SUM(O.QUANTITY * O.PRICE) AS TOTAL_REVENUE
    FROM PRODUCTS P
    JOIN ORDER_ITEMS O
    ON P.PRODUCT_ID = O.PRODUCT_ID
    GROUP BY
        P.CATEGORY_ID,
        P.PRODUCT_NAME
),
CTE2 AS
(
    SELECT
        *,
        ROW_NUMBER() OVER
        (
            PARTITION BY CATEGORY_ID
            ORDER BY TOTAL_REVENUE DESC
        ) AS RN
    FROM CTE
)
SELECT
    CATEGORY_ID,
    PRODUCT_NAME,
    TOTAL_REVENUE
FROM CTE2
WHERE RN = 1

```

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9.SELECT EMAIL
FROM EMPLOYEES
GROUP BY EMAIL
HAVING COUNT(*) > 1

```

```
or
WITH CTE AS
(
    SELECT EMAIL
    FROM EMPLOYEES
    GROUP BY EMAIL
    HAVING COUNT(*) > 1
)
SELECT EMAIL
FROM CTE;
```

```
10.SELECT
    T1.CUSTOMER_ID,
    T1.TRANSACTION_ID AS TRANSACTION_1,
    T1.CITY AS CITY_1,
    T1.TRANSACTION_TIME AS TIME_1,
    T2.TRANSACTION_ID AS TRANSACTION_2,
    T2.CITY AS CITY_2,
    T2.TRANSACTION_TIME AS TIME_2
FROM TRANSACTIONS T1
JOIN TRANSACTIONS T2
    ON T1.CUSTOMER_ID = T2.CUSTOMER_ID
    AND T1.TRANSACTION_ID < T2.TRANSACTION_ID
    AND T1.CITY <> T2.CITY
    AND T1.TRANSACTION_TIME = T2.TRANSACTION_TIME;
```